

Composing composite size morphemes*

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1. Introduction

In this paper, I argue that Brazilian Portuguese (BP) diminutive and augmentative formation (henceforth, size formation) involves syntactic compounding, even though size words are prosodic words formed by affixation.¹ More specifically, size morphemes attach to bases by adjunction or incorporation and are themselves syntactically complex, consisting of a root merged with a category-defining head, e.g., $[\sqrt{\text{DIM}}\ n]_n$. I further argue that the analysis supports Harðarson (2016)'s Matching Condition on compounding.

2. Size word formation – the empirical landscape

BP size words are formed by attaching the diminutive suffix *-inh* or the augmentative suffixes *-ão* (MASC) and *-on* (FEM) to either a themeless stem or an inflected word (1,2).

(1) Themeless stem base	(2) Word base
a. livr-inh-o book-DIM-TVI 'small book; big book'	a. livr-o-z-inh-o book-TVI-EPEN-DIM-TVI 'small book'
b. livr-ão book-AUG.M 'big book'	b. livr-o-z-ão book-TVI-EPEN-AUG.M 'big book'

They attach to bases of several lexical categories, except for verbal stems (3).

*I thank people from UConn Linguistics, especially Andrea Calabrese, Adrian Stegovec, Vicki Carstens, and Qiushi Chen, for insightful comments; Jonathan Bobaljik and Maria Gouskova for valuable feedback; the NELS 56 audience for engaging discussion; and the proceedings editors for their work.

¹For BP size morphology, see, e.g., Menuzzi (1993), Bachrach and Wagner (2007), Ferreira (2005), Bisol (2010), Armelin (2014), Barbosa and Freitas (2014), Rocha and Vicente (2016), Ulrich and Schwindt (2018). For overviews, see Lee (2013) and Newell and Eddington (2023).

- (3) a. livro (N); livrinho/livão
book book.DIM/AUG
- b. alto (A); altinho/altão
tall tall.DIM/AUG
- c. cedo (ADV); cedinho/cedão
early early.DIM/AUG
- d. amar (V); *aminhar/*amão
love.INF love.DIM/AUG

The resulting theme vowel may differ from that of the base stem (4).

- (4) pent-e; pent-inh-o; pent-e-z-inh-o
comb-TVIII comb-DIM-TVI comb-TVIII-EPEN-DIM-TVI
'comb; little comb; little comb'

Size-word bases are also opaque to phonological processes typically found in BP derivation (5). Note that the diminutive, contrary to the nominalizer, does not trigger root allomorphy.

- (5) 'b[ɔ].l-a; b[o]'l-a.d-a; 'b[ɔ].l-inh-a
ball-TVII ball-NMZR-TVII ball-DIM-TVII
'ball; ball hitting; little ball'

Base size also constrains interpretation: stem-based but not word-based formation permits noncompositional (idiomatic) readings (6).

- (6) camis-a; camis-inh-a; camis-a-z-inh-a
shirt-TVII shirt-DIM-TVII shirt-TVII-EPEN-DIM-TVII
'shirt; small shirt/condom; small shirt'

Lastly, stem-based augmentatives show a gender asymmetry: masculine *-ão* may override base gender and allow idiomatic readings, but feminine *-on* is always compositional (7).

- (7) mochil-a (F); mochil-ão (M); mochil-on-a (F)
backpack-TVII; backpack-AUG.M; backpack-AUG.F-TVII
'backpack; big backpack/backpacking (trip); big backpack'

3. BP size morphemes are adjuncts (except when they're not)

BP size morphemes have been analyzed as heads (Rosa 1982, Armelin 2014), as in Russian (Steriopolo 2009), or as adjuncts (Bachrach and Wagner 2007), as in Italian (Scalise 1988). I reconcile these views by proposing that their status depends on how they attach to the base.

As adjuncts, they preserve the base's category (3) and gender (8) and may attach to either stems or words (9).

- (8) a. livro (M), livrinho/livrozinho (M); planeta (M), planetinha/planetazinho (M)
book book.DIM planet planet.DIM

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- b. revista (F), revistinha/revistazinha (F); foto (F), fotinha/fotozinha (F)
 magazine magazine.DIM photograph photograph.DIM
- (9) livr-inho, livro#zinho; revist-inha, revist-a#zinha
 book-INHO book+TVI#ZINHO magazine-INHA magazine-TVII#ZINHA

As heads, their behavior is more restricted, *quasi*-exclusive of the augmentative masculine -ão. Note that they may change the lexical category (10) or gender (7) of the base.

- (10) a. bunda (N); bund-ão (A) b. empurrar (V); empurr-ão (N)
 butt.F butt.AUG.M push push-AUG.M
 ‘butt; coward’ ‘(to) push; big push’

Another restriction is that SIZE is a head only when is attached to a themeless stem (11). In other words, word-based formation must always involve adjunction. It can’t therefore change the lexical category or gender of the base, and it also disallows an idiomatic reading.

- (11) *bunda-zão, #bunda-zona
 butt.F-AUG.M butt.F-AUG.F
 (only okay as the noun “big butt”)

4. But adjuncts to what?

At an observational level, SIZE seems to attach to a themeless stem (1), or to a prosodic (inflected) word (7). This can actually be shown to be the adequate description. Regarding word attachment, note that plural nominals (12), nominal verbal forms (13), and even tensed verbs (14), all prosodic inflected words, may serve as the base for size words.²

- (12) a. leão; leõe-s; leõe-z-inh-o-s; leõe-z-õe-s
 $\sqrt{\text{lion}}$ $\sqrt{\text{lion}}$ -PL $\sqrt{\text{lion}}$.PL-EPEN-DIM-TVI-PL $\sqrt{\text{lion}}$.PL-EPEN-AUG.PL-PL
- b. senhor senhore-s; senhore-z-inh-o-s; senhore-z-õe-s
 $\sqrt{\text{gentleman}}$ $\sqrt{\text{ }}$ -PL $\sqrt{\text{ }}$ -EPEN-DIM-TVI-PL $\sqrt{\text{ }}$ -EPEN-AUG.PL-PL
- c. pap[ɛʊ]; pap[ɛɪ]-s; pap[ɛɪ]-z-inh-o-s; pap[ɛɪ]-z-õe-s
 $\sqrt{\text{paper}}$ $\sqrt{\text{paper}}$ -PL $\sqrt{\text{paper}}$ -EPEN-DIM-TVI-PL $\sqrt{\text{ }}$ -EPEN-AUG.PL-PL
- (13) a. am-a-r; am-a-r-z-inh-o
 $\sqrt{\text{love}}$ -VTV-INF $\sqrt{\text{love}}$ -VTV-INF-EPEN-DIM-TVI
 ‘(to) love; (to) love a little’
- b. am-a-d-o; am-a-d-inh-o; am-ad-o-z-inh-o
 $\sqrt{\text{love}}$ -VTV-PART-TVI $\sqrt{\text{love}}$ -VTV-PART-DIM-TVI

²Note that the consonant /z/ is epenthetically inserted to avoid hiatus (Bisol 2010).

‘loved; loved a little’

- c. am-a-nd-o; am-a-nd-inh-o; am-a-nd-o-z-inh-o
 $\sqrt{\text{love}}$ -VTV-GER-TVI; $\sqrt{\text{love}}$ -VTV-GER-DIM-TVI
 ‘loving; loving a little’

- (14) am-ei; am-ei-z-inh-o
 $\sqrt{\text{love}}$ -VTV-1.SG.PST.IND $\sqrt{\text{love}}$ -VTV-1.SG.PST.IND-EPENDIM-TVI
 ‘(I) loved; (I) loved a little’

Now, what is the syntactic status of a “themeless stem”? Is it a bare root ($\sqrt{\text{CAT}}$) or a categorized root ($[\sqrt{\text{CAT}}\ n]_n$)? I argue for the latter. First note that, although size morphemes may adjoin to several lexical categories, they are strictly disallowed with verbal stems (15).

- (15) a. am-a-r; *am-inh-a-r
 love-VTV-INF love-DIM-VTV-INF
 ‘(to) love; (to) love a little (int.)’
 b. beb-e-r; *beb-inh-e-r
 drink-VTV-INF drink-DIM-VTV-INF
 ‘(to) drink; (to) drink a little (int.)’

Crucially, the bare root view overgenerates. Considering that roots are category-neutral,³ direct root attachment would wrongly predict the starred verbal forms in (15).

I therefore propose that size morphemes require at least one level of functional material on the base they head or adjoin to. There is actually evidence for this claim. Note that the augmentative word in (16) actually comes from the verb: the fact that the root final consonant is palatalized suggests the underlying presence of the verbal thematic vowel, given the former is known to palatalize preceding dorsals in some varieties of Romance (A. Calabrese, p.c.). Crucially, the verbal theme vowel would diagnose the presence of a verbal domain (Oltra-Massuet 1999, Calabrese 2023).

- (16) fu[ʒ]ir (V); fu[g]a (N); fu[ʒ]-ão (N); *fu[g]-ão
 escape_{.Inf} escape escaper (someone who escapes a lot)

5. Composite size morphemes

I analyze SIZE as a root merged with a categorizer, rather than an atomic categorizer or a bare root.⁴ First, I show that SIZE is not an atomic category defining functional node such as *n*, *a*, *adv*, i.e., SIZE is not a derivational morpheme in the traditional sense.⁵

³See, e.g., Harley (2005), Embick (2010), Borer (2014).

⁴See Gouskova and Bobaljik (2022) for discussion of complex affixes, including diminutives.

⁵I assume that derivational morphemes are category defining heads selecting for an uncategorized root; see, e.g., see Marantz (2001), Arad (2003), Harley (2005), Embick and Noyer (2007). See also Creemers et al. (2018) for a different view.

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Regarding adjunction, size morphemes behave differently from typical cases of derivational morphology in BP. In this respect, they don't control the lexical category or the gender output (8), are not picky about the category of the base (3), may occur after inflectional morphology (12). They are also known for bleeding phonological processes otherwise attested (17), and are relatively more productive compared to derivational morphemes (18).

- (17) a. 'b[ɔ].l-a; b[o]'l-a.d-a; b[ɔ]'l-i.nh-a
 ball-TVII ball-NMZR-TVII ball-DIM-TVII
 'ball; ball hitting; small ball'
- b. 'f[ã]m-a; f[a]'m-os-o; f[ã]'m-inh-a
 fame-TVII fame-NMZR-TVI fame-DIM-TVII
 'fame; famous; a bit of fame'
- (18) livr-aria, ?revist-aria; alt-ur-a, ?baix-ur-a
 book-NMZR magazine-NMZR tall-NMZR-TVII short-NMZR-TVII

Secondly, I show that they are also not bare roots ($\sqrt{\text{DIM}}$ or $\sqrt{\text{AUG}}$). In this respect, they simply don't distribute as roots (19, 20) (Take 'b' as reactions to 'a').

- | | |
|--|---|
| <p>(19) a. Comprei um sofá pequeno.
 bought a couch small
 'I bought a small couch.'</p> <p>b. Que pequen-íssim-o!
 how small-SPRLTV-TVI
 'How small!'</p> | <p>(20) a. Comprei um sofazinho.
 bought a couch.DIM
 'I bought a small couch.'</p> <p>b. *Que (z)inh-íssim-o!
 how DIM-SPRLTV-TVI
 'How small!' (Intended)</p> |
|--|---|

Considering roots are syntactically inert (Acquaviva 2009, Alexiadou 2014), they are predicted to not exhibit either *head* or *adjunct* behavior. Size morphemes, however, do. I conclude SIZE morphemes are not purely functional heads or simply bare roots, and propose that they correspond, instead, to a root merged with a category defining functional head. This allows us to account for seeming contradictions regarding their behavior. We saw cases where size morphemes behaves like heads. Regarding their root behavior, note that masculine AUG *-ão* undergoes the same type of allomorphy that plural triggers on roots (21).

- (21) leão; leõe-s; leõe-z-inh-o-s; leõe-z-õe-s
 lion lion-PL lion.PL-EPEN-DIM-TVI-PL lion.PL-EPEN-AUG.PL-PL

In the following, I propose the structures corresponding to the range of BP size words, showing how they account for the array or properties discussed above. Crucially, the analysis advocates for their compound status.

(22) a. pr^on^to#soc^orro → pr^on^t[u]#soc^orr[u]; cidáde#satélite → cidád[i]#satélit[i]
ready#help city#satellite
'emergency room; satellite city'

(23) água oxigenáda → àgu[o]xigenáda; água → agu[i]nha
water oxygenated water water.DIM Guzzo (2018:698-9)

(25) $\text{súper\#mercádo} \rightarrow \text{super\#market}$ $\text{ácido-zinho} \rightarrow \text{acid-DIM}$ Guzzo (2018:704-5)

(26) chap[ɛ]u; chap[ɛ]l-aria; chap[ɛ]u-zinho; chap[ɛ]u#coco
 hat hat-NMZR hat-DIM hat#coconut
 ‘hat; hat store; small hat; bowler hat’ Guzzo (2018:706)

(27) a. linda-mente e suave-mente → linda- $\emptyset$ e suave-mente
beautiful-ly and soft-ly beautiful- $\emptyset$ and soft-ly

⁶See Guzzo (2010), Silva (2010), Guzzo (2018), Toneli (2014).

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- b. pré-cirurgia e pós-cirurgia → pré-∅ e pós-cirurgia
 pre-surgery and post-surgery pre-∅ and post-surgery

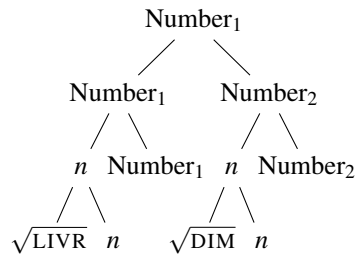
Finally, they may also display word-like independent morphosyntactic behavior (28) (28b in ungrammatical out of the blue)

- (28) a. Olha aquele cachorr-inho!
 look that dog-DIM
 ‘Look at that little dog!’
 b. Bem inho mesmo!
 well DIM indeed
 ‘It is indeed very small!’

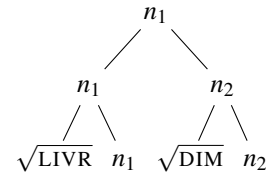
In view of the discussion above, I propose an account for the formation of BP size words. They come in different sizes, depending on (i) how size morphemes attach to the base (via adjunction or via head incorporation), and on (ii) the base they attach to, which may refer to a (prosodic) word or a (themeless) stem.

Adjunction structures A size morpheme may either adjoin to a prosodic word (29), or to a themeless stem (30), as represented below for *livrozinho* and *livrinho*, respectively.

- (29) Word-based size word

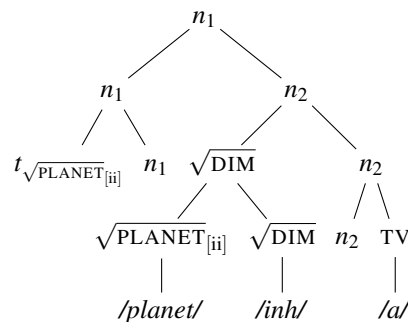


- (30) Stem-based size word formation



Stem-based incorporation In this derivation, a size morpheme, in the status of a head, selects for a noun, i.e., a root categorized by *n*. I assume that selection triggers incorporation in this case, thus giving rise to the size word structure in (31).

- (31) Stem-based size word (via incorporation)



The incorporation analysis allows us to account for the preservation of the marked theme vowel *-a* “across” the size morpheme in *planet-inh-a* (‘small planet’), from *planet-a* (‘planet’), both masculine words. Assuming that the realization of marked theme vowels is controlled by a diacritic on the root, incorporation of the bare root *planet* would include the diacritic in a domain where it is local to the theme vowel node, thus accounting for the preservation of the marked vowel.

The analysis above can be taken to provide support for Harðarson (2016)’s Matching Condition on the formation of compounds, which states that members of a compound must be of the same size and type. Note that the structures above do conform to this constraint. There is also empirical support for this constraint. Consider (32).

- (32) flor-z-inh-a-s; flore-z-inh-a-{s/*Ø}
 $\sqrt{\textit{flower}}$ -EPEN-DIM-TVII-PL $\sqrt{\textit{flower}}$.PL-EPEN-DIM-TVII{-PL/*-Ø}
 ‘small flowers’

(32) shows that the outer plural must be realized whenever the inner plural is realized. I take this to be a Matching Condition effect: when the size morpheme on the ungrammatical word (**florezinha*-Ø) lacks their outer # node, it therefore cannot attach to a plural base.

7. Selective idiomaticity

Remember that only stem-based formation permits idiomatic readings (6); cf. Armelin (2014). If non-compositional meaning is fixed in the first root–categorizer domain (Arad 2003), word-based SIZE then attaches after the base meaning has been fixed, whereas stem-based compounding places both roots within the relevant domain.

8. Conclusion

In this paper I analyzed diminutive and augmentative formation in Brazilian Portuguese. I argued that BP size words are compounds whose complex size morpheme, [$\sqrt{\textit{SIZE}}$ *n*]_n, attaches to a categorized stem or inflected word by adjunction or incorporation. This analysis unifies their head/adjunct behavior, supports the view that affixes may be internally complex (Gouskova and Bobaljik 2022), and derives Harðarson (2016)’s requirement that compound members match in structural size and type.

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